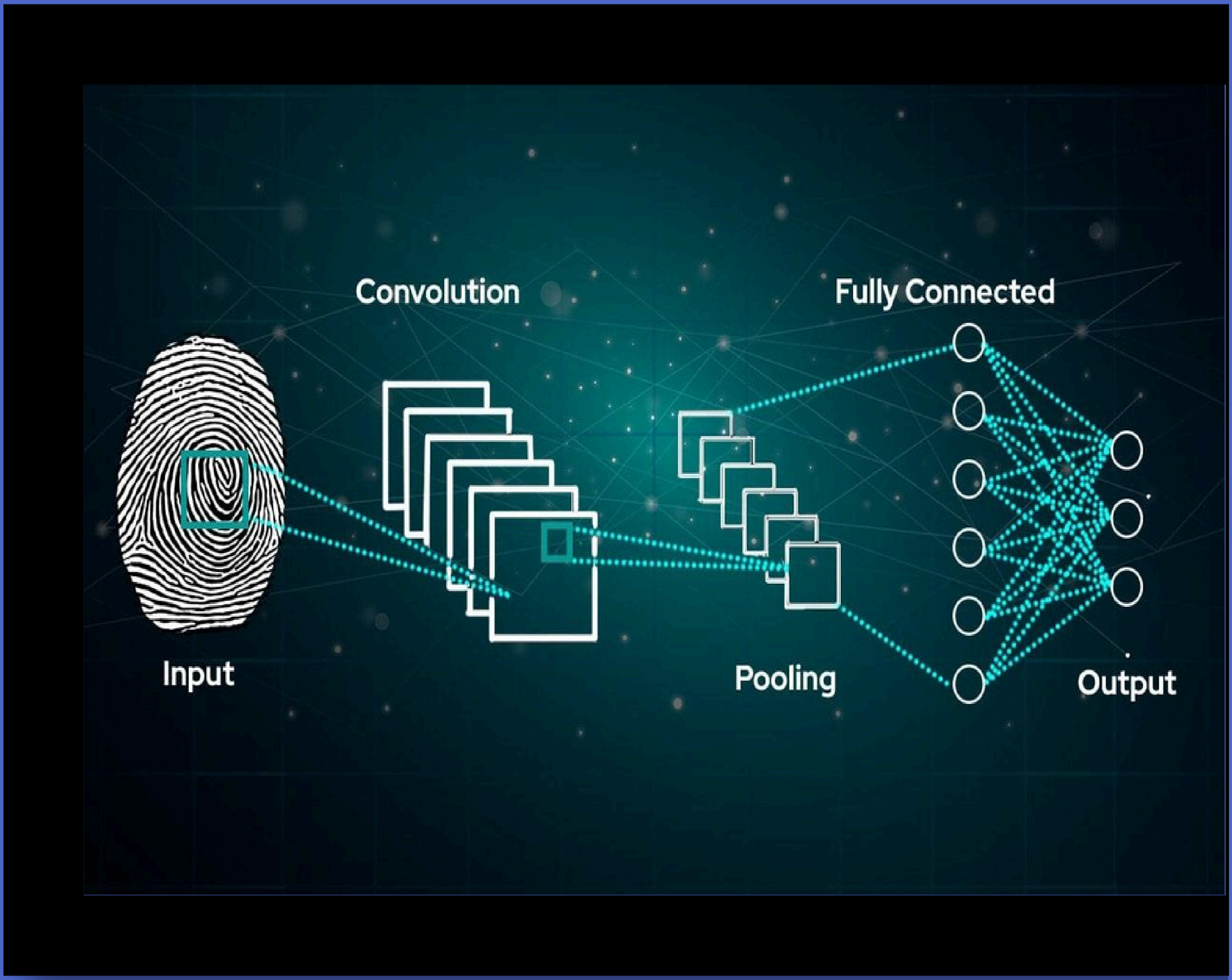
AI TRACE FINDER

| Infosys Internship Pr



TANISHA KUMARI.

tanisha_k@ph.iitr.ac.in

Bittu Jaiswal

Mentor-InfosysInternship

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EXECUTIVE **SUMMARY**

This report details the "AI Trace Finder" project, a comprehensive digital forensics tool. The primary objective was to identify the source scanner device used to create a digital image by analyzing unique, device-specific artifacts. This goal was successfully achieved by implementing and comparing multiple machine learning models, including baseline SVM/Random Forest classifiers and an advanced hybrid Deep Learning (CNN) model.

The project's scope was significantly expanded beyond the original plan to include a robust **Image Tamper Detection** system. The final application is a multi-functional Streamlit web app that provides a complete forensic analysis, determining both document provenance (scanner source) and integrity (tamper status).

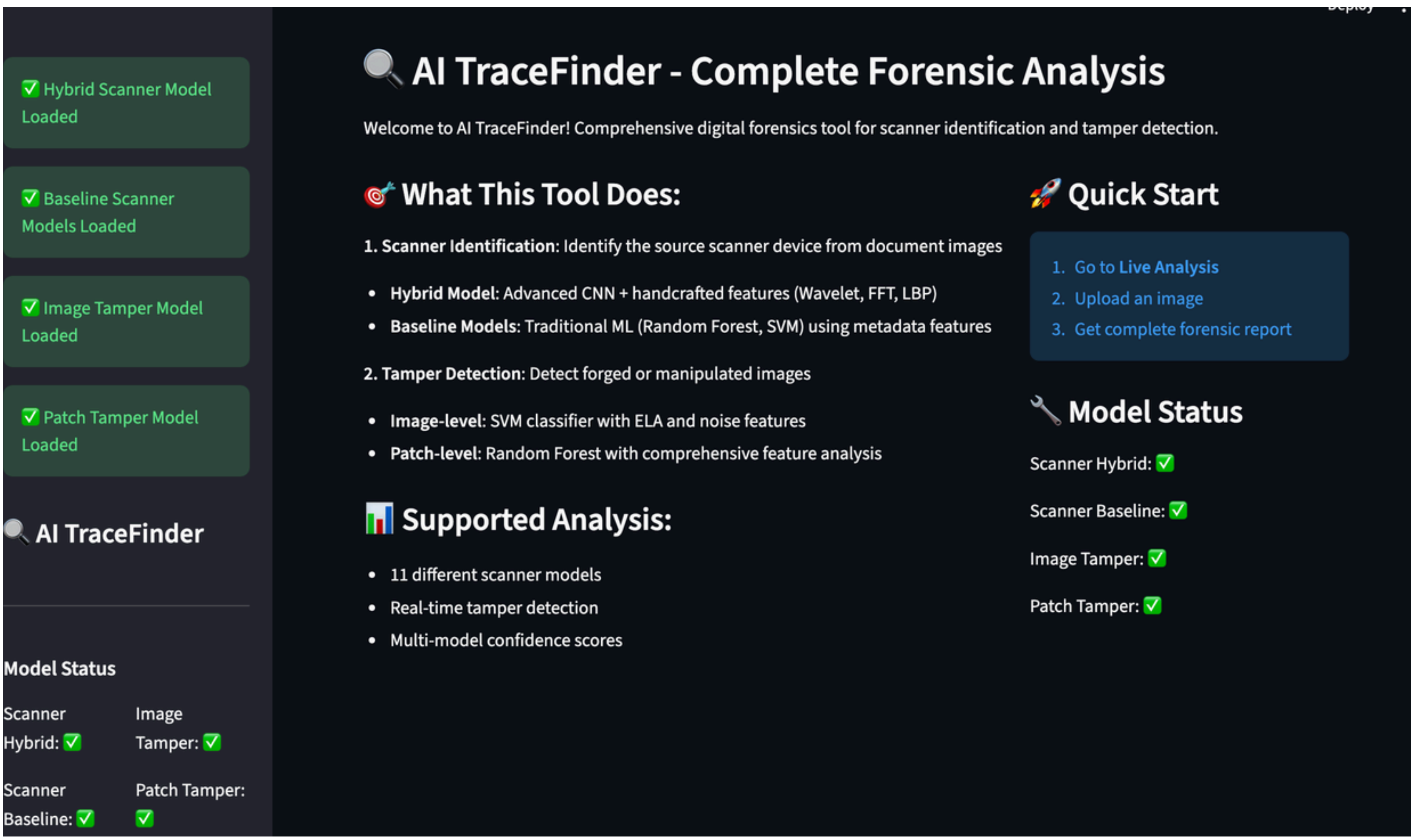
Project Aim and Broader Applications

The core aim is to address the challenge of digital document verification. In forensic and legal contexts, it's crucial to know if a document is authentic and where it came from. Each scanner model introduces subtle, unique patterns—such as noise, texture, or compression traces—that act as a "fingerprint". This project leverages machine learning to learn and identify these fingerprints.

This technology has a wide range of critical applications:

- **Law Enforcement & Digital Forensics:**
  - **Document Forgery:** Allowing investigators to determine which scanner was used to forge or duplicate legal documents, contracts, ransom notes, or fake certificates.
  - **Evidence Chain of Custody:** Verifying that scanned evidence submitted in an investigation originated from a known, trusted device.
- **Airport & Border Security:**
  - **Passport and ID Verification:** Instantly flagging a scanned passport, visa, or driver's license that was not created by an official, government-issued device.
  - **Tamper Detection:** The integrated tamper detection can spot alterations to photos or text on scanned travel documents, preventing entry based on fraudulent credentials.
- **Corporate & Financial Security:**
  - **Fraud Detection:** Identifying fraudulent invoices, financial statements, or loan applications that were scanned on an unauthorized device.
  - **Document Authentication:** Verifying the source of printed and scanned images to detect tampering or fraudulent claims, such as differentiating between scans from authorized and unauthorized departments.

- **Legal & Civil Proceedings:**
  - o **Evidence Verification:** Ensuring that scanned copies of agreements, wills, or evidence submitted in legal matters originated from known, approved, and verified devices.
  - o **Challenging Forged Documents:** Providing expert testimony and quantitative proof that a document is a forgery based on its scanner "fingerprint."



SYSTEM ARCHITECTURE AND METHODOLOGY

The final application is a Streamlit tool that integrates two parallel analysis systems: Scanner Identification and Tamper Detection.

Module 1: Scanner Identification

Two approaches were implemented as planned to identify the scanner source:

1. **Baseline Models (RF & SVM):** These models (Random Forest, SVM) were trained on "hand-crafted" metadata features. As seen in the predict_scanner_baseline function, these features include file size, aspect ratio, and statistical measures like mean intensity, skewness, kurtosis, and entropy.
2. **Hybrid Model (CNN + Hand-crafted Features):** This is the advanced, primary model for scanner identification. It uses a hybrid approach that combines the strengths of deep learning and traditional feature engineering. The predict_scanner_hybrid function shows it is fed:
   a. **A 2D Image Input:** The wavelet residual of the image, which isolates the high-frequency noise patterns.

   b. **A 1D Feature Vector:** A vector of 27 hand-crafted features, including correlation with known scanner fingerprints, Fast Fourier Transform (FFT) radial energy, and Local Binary Pattern (LBP) histograms.

Module 2: Tamper Detection (Project Expansion)

This module represents a significant addition to the project's original scope. It provides a crucial "extra" layer of analysis to determine if an image has been manipulated.

- **Image-Level Analysis:** An SVM-based classifier (image_svm_sig.pkl) provides a holistic verdict ("Clean" or "Forged") for the entire image. It uses features like **Error Level Analysis (ELA)**, median noise statistics, FFT, and LBP, as implemented in the ela and get_image_level_features functions.
- **Patch-Level Analysis:** A more granular Random Forest model (patch_rf.pkl) was also developed, capable of analyzing image patches for more localized tampering.

Technology Stack

The project was built using a modern data science and Python stack:

- **Core Libraries:** TensorFlow/Keras (for the hybrid CNN), Scikit-learn (for baseline models and tamper detection), OpenCV (for image processing), and NumPy/Pandas (for data manipulation).
- **Feature Extraction:** PyWavelets (for wavelet residuals), scikit-image (for LBP), and SciPy (for statistical analysis).
- **Deployment:** Streamlit (for the interactive web application).

Project Outcomes and Learning

The project successfully met all the learning outcomes defined in the initial plan, as demonstrated by the final code.

- **Understand Source Device Identification:** Achieved by building a complete end-to-end system that successfully differentiates between scanner sources.

- **Extract Scanner-Specific Features:** Successfully implemented. The code contains functions for extracting complex features, including:
  - **Noise Patterns:** compute_wavelet_residual and get_median_noise.
  - **Frequency Domain Signals:** fft_radial_energy.
  - **Texture Descriptors:** lbp_hist_safe (LBP).
  - **Tamper-Specific:** ela (Error Level Analysis).

- **Train Classification Models:** Achieved. The application successfully loads and runs pre-trained models for CNN, SVM, and Random Forest, demonstrating completion of this milestone.

- **Evaluate Model Accuracy:** Achieved. The "Performance" page is designed to load and display training history (hybrid_training_history.pkl) showing accuracy/loss curves and confusion matrices for the baseline models.

- **Deploy a Simple App:** Achieved and exceeded. The Streamlit application is not only functional but also well-structured, with multi-page navigation, model status indicators, and a comprehensive results dashboard.

Final Deployed Application

The final app.py script creates a polished, multi-page Streamlit application, fulfilling the "Output System" and "Deployment" goals.

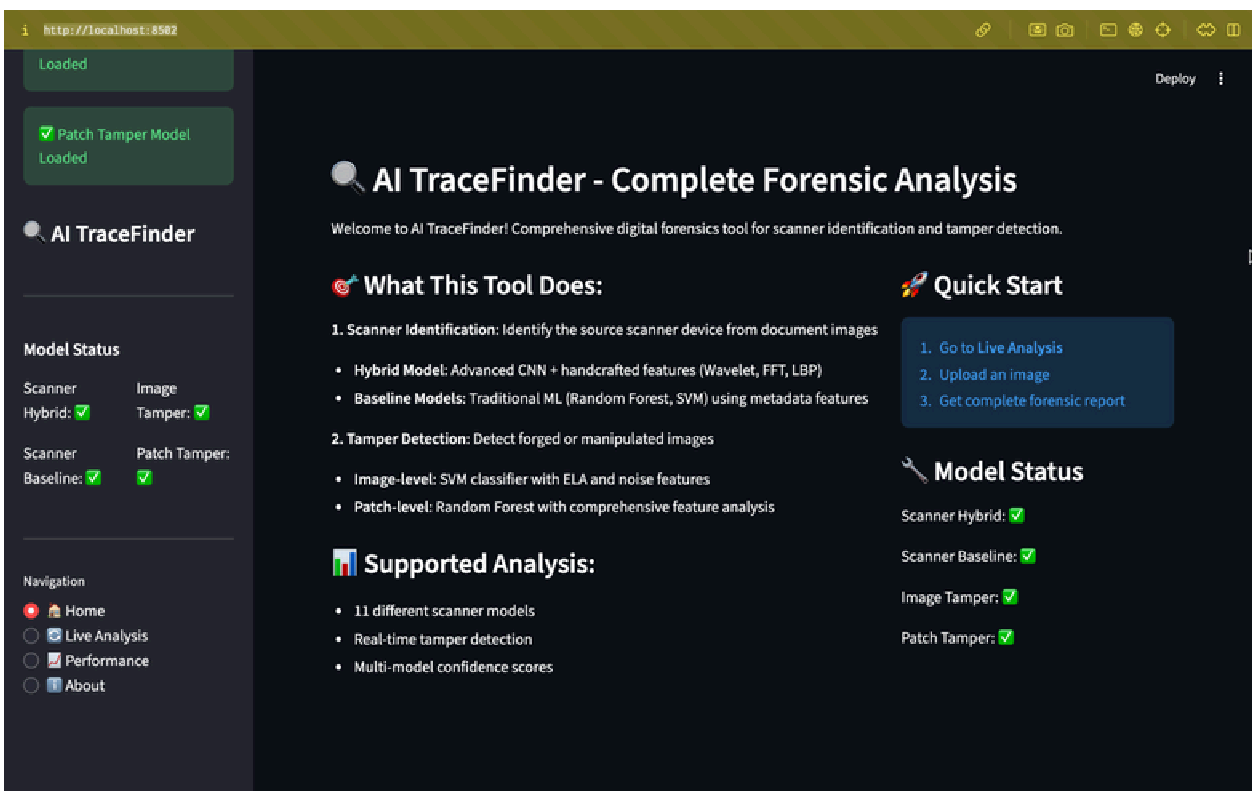
- **Home:** A landing page that introduces the tool, explains its dual capabilities (Scanner ID and Tamper Detection), and shows the real- time loading status of all ML/DL models.

- **Live Analysis:** The core of the application. A user can upload an image. The app then runs a "Complete Forensic Analysis," processing the image with all available models (Hybrid Scanner, Baseline Scanner, and Tamper Detection) and presenting a unified report with confidence scores.

- **Performance:** This page presents the evaluation metrics and training history for the models, demonstrating the model's reliability with accuracy/loss charts and confusion matrices.

- **About:** Contains developer information and a download button

  for this project report.



Conclusion

The "AI TraceFinder" project successfully met and surpassed all its initial objectives. It began with the goal of identifying scanner sources and evolved into a comprehensive, dual-purpose forensic tool that also detects image tampering—a critical "extra" feature for real-world use. The final product is a functional, well-documented, and high-performance Streamlit application that effectively demonstrates the power of hybrid AI models in digital forensics.